# Preparation of 0.1 M Acetate Buffer (pH 5.0)

## Aim

To prepare 100 mL of 0.1 M acetate buffer of pH 5.0 using 0.1 M sodium acetate and 0.1 M acetic acid solutions.

## Required Solutions

- 0.1 M Acetic acid solution ($CH_3COOH$)
- 0.1 M Sodium acetate solution ($CH_3COONa$)
- Distilled water ($H_2O$)

## Apparatus

- 100 mL volumetric flask
- measuring cylinders
- Beaker
- Glass rod
- pH meter

## Theory

An acetate buffer consists of a weak acid (acetic acid, $CH_3COOH$) and its conjugate base (sodium acetate, $CH_3COONa$). It resists changes in pH when small amounts of acid or base are added. The pH of the buffer is governed by the Henderson–Hasselbalch equation:

$$\mathrm{pH} = \mathrm{p}K_a + \log\left(\frac{[\mathrm{CH_3COO^-}]}{[\mathrm{CH_3COOH}]}\right)$$

For acetic acid,

$$\mathrm{p}K_a = 4.76$$

For a buffer of pH 5.0,

$$5.0 = 4.76 + \log\left(\frac{[CH_3COO^-]}{[CH_3COOH]}\right)$$

Therefore,

$$\frac{[CH_3COO^-]}{[CH_3COOH]} = 10^{(5.0-4.76)} = 10^{0.24} \approx 1.74$$

Since both stock solutions have the same concentration (0.1 M), the concentration ratio is equal to the volume ratio.

Thus,

$$\frac{V_{CH_3COONa}}{V_{CH_3COOH}} = 1.74$$

For a total volume of 100 mL,

$$V_{CH_3COONa} = \frac{1.74}{1+1.74} \times 100 = 63.5 \text{ mL}$$

$$V_{CH_3COOH} = \frac{1}{1+1.74} \times 100 = 36.5 \text{ mL}$$

Hence, mixing 63.5 mL of 0.1 M sodium acetate with 36.5 mL of 0.1 M acetic acid produces a 0.1 M acetate buffer of pH 5.0.

# Procedure

1. Clean a 100 mL volumetric flask and rinse it with distilled water.
2. Measure 63.5 mL of 0.1 M sodium acetate ($CH_3COONa$) using measuring cylinder and add into the volumetric flask.
3. Add 36.5 mL of 0.1 M acetic acid ($CH_3COOH$) into above solution.
4. Mix the solution thoroughly with constant stirring.
5. Measure the pH using a calibrated pH meter.
6. If necessary, adjust the pH:
   - Add a few drops of 0.1 M acetic acid to decrease the pH.
   - Add a few drops of 0.1 M sodium acetate to increase the pH.
7. If total volume is slightly less than 100 mL after pH adjustment, make up the volume to mark with distilled water.

# Observation

The prepared solution was clear and colourless. The measured pH was approximately 5.0.

# Result

A 100 mL sample of 0.1 M acetate buffer of pH 5.0 was successfully prepared by mixing 63.5 mL of 0.1 M sodium acetate solution with 36.5 mL of 0.1 M acetic acid solution.

# Precautions

1. Wash the Apparatus carefully.
2. pH meter must be calibrated before use.
3. Handle Acid solution with care.